AN ARTIN-REES CONSTANT SURVIVES A FLAT BASE CHANGE OF NOETHERIAN RINGS

THE STACKS PROJECT AUTHORS

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Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The Artin-Rees lemma and the meaning of a constant c that works for a map of finite modules. stated and discussed immediately before this lemma, source lines 194-282. Supplies the hypothesis on c over A and the form the conclusion has to take over B.
- Flatness of A over B makes tensoring with B commute with the formation of kernels. used at proof lines 301-303 to identify the preimage of the extended ideal with the tensor product of the preimage. Is the only place where the flatness hypothesis is used, and it is what makes the base change computation possible.
- The preimage of an ideal under a map of modules is computed by the kernel of the induced map to the quotient. used at proof lines 296-300. Rewrites the preimage in the two forms whose comparison produces the result.

Source reference. Original lemma 07VG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10006–10006. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let $A \to B$ be a flat map of Noetherian rings. Let $I \subset A$ be an ideal. Let $f : M \to N$ be a homomorphism of finite A-modules. Assume that c works for f in the Artin-Rees lemma. Then c works for $f \otimes 1 : M \otimes_A B \to N \otimes_A B$ in the Artin-Rees lemma for the ideal IB.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Note that

$$(f \otimes 1)(M) \cap I^n N \otimes_A B = (f \otimes 1)\left((f \otimes 1)^{-1}(I^n N \otimes_A B)\right)$$

On the other hand,

$$\begin{aligned}(f \otimes 1)^{-1}(I^n N \otimes_A B) &= \mathrm{Ker}(M \otimes_A B \to N \otimes_A B/(I^n N \otimes_A B)) \\ &= \mathrm{Ker}(M \otimes_A B \to (N/I^n N) \otimes_A B)\end{aligned}$$

As $A \to B$ is flat taking kernels and cokernels commutes with tensoring with B, whence this is equal to $f^{-1}(I^n N) \otimes_A B$. By assumption $f^{-1}(I^n N)$ is contained in $\mathrm{Ker}(f) + I^{n-c} M$. Thus the lemma holds. $\square$